INCH-POUND

MIL-B-70838 (AR)
19 May 1989

MILITARY SPECIFICATION

BILLET, DEPLETED URANIUM

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements, quality assurance provisions and preparation for delivery of depleted uranium billets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9650

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2.1.2 Other Government documents and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

Title 49 (49 CFR)	Hazardous Materials Regulations of the Department of Transportation

(Applications for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 8	Tension Testing of Metallic Materials, Methods of.
ASTM E 18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test Methods for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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See para. 1.2

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Billets may be manufactured from either virgin derby material, recyclable scrap material or a combination thereof.

3.1.2 Records. The contractor shall maintain records of the charge composition for each casting lot. These records shall specify, by weight, type and source of supply the amount of virgin derby and recyclable scrap (see 6.4) contained within each casting lot. Recyclable scrap shall be categorized as: top crop, sectioned billet or sectioned plate.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2.1 Mold wall identification. For first article only, the sides of the billet will be identifiable to the orientation of the billet within the mold.

3.2.2 Macrostructure. Macrostructure specimens shall be fabricated and photomacrographs made from the first article sample in accordance with 4.5.1.5.1 for information only.

3.2.3 Microstructure. Microstructure specimens shall be fabricated and photomicrographs made from the first article sample in accordance with 4.5.1.5.2 for information only.

3.3 Traceability. Each billet shall be identified by a unique serial number/code which is traceable to the chemical analysis (see 3.4), hydrogen content (see 3.5), and casting parameters (see 3.12) for the casting lot from which the billet is drawn and any rework or repair operations (see 6.14) performed on that billet. The contractor's traceability plan shall be submitted to the Government for review in accordance with DI-R-5297/T (see 6.2.1 and 6.2.2.2).

3.3.1 Marking. Each billet shall be permanently marked or stamped with the unique identification number/code assigned to that billet. This marking shall be clearly and easily legible and shall be on the top surface (after cropping) of the billet. The contractor's proposed method for marking of billets shall be submitted to the Government for review in accordance with DI-P-1604/T (see 6.2.1 and 6.2.2.3).

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3.3.2 Records. Unless otherwise specified, traceability and inspection data in accordance with UDI-R-22706A (see 6.2.1) shall be included with each shipment for the billets included in that shipment. In addition, the contractor shall maintain traceability and inspection and test records for a period of not less than five years from the completion of the contract or purchase order.

3.4 Chemistry. Billets shall exhibit the following chemistry when analyzed in accordance with 4.5.1.2 or 4.5.2.2:

Uranium^{235} (U^{235})	0.25	weight percent maximum
Uranium (U)	99.95	weight percent minimum
Carbon (C)	120	parts per million maximum
Nickel (Ni)	80	parts per million maximum
Iron (Fe)	70	parts per million maximum
Zirconium (Zr)	60	parts per million maximum
Silicon (Si)	50	parts per million maximum
Titanium (Ti)	30	parts per million maximum
Aluminum (Al)	20	parts per million maximum
Other elements	100	parts per million maximum

3.4.1 Uranium content. Combined total of Uranium^{235} and Uranium^{238} which may be determined by difference.

3.4.2 Other elements. Other elements to be analyzed for and reported are: Beryllium (Be), Bismuth (Bi), Boron (B), Cadmium (Cd), Calcium (Ca), Chromium (Cr), Copper (Cu), Lead (Pb), Magnesium (Mg), Manganese (Mn), Molybdenum (Mo), Niobium (Nb), Sodium (Na), Strontium (Sr), Tin (Sn), Vanadium (V), and Zinc (Zn).

3.5 Hydrogen. The hydrogen content of the billet shall be less than 1.0 part per million when analyzed in accordance with 4.5.1.1 or 4.5.2.1.

3.6 Matrix inclusions. Billets shall be free from matrix inclusions larger than 0.0008 inch (0.080 inch at 100X) with a concentration not to exceed 11 inclusions per 0.02-inch square (2-inch square at 100X).

3.7 Configuration. The dimensions and configuration of the billet shall be as specified in the contract or purchase order (see 6.2.c).

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described herein shall consist of this document and

3.8 Subsequent processing capability. The billet shall be free of defects which cause any of the following type defects in plate rolled from the billet:

a. Cracks;
b. Holes;
c. Surface discontinuities (e.g., indentations) greater than 0.015 inch in depth and 1.0 inch in length.

The contractor shall establish criteria for assuring that each billet meets the requirements specified above. This criteria shall apply to both the first article sample and to production billets and shall be submitted to the Government for review in accordance with DI-R-5297/T (see 6.2.1 and 6.2.2.4). Any conflict as to the acceptability of the proposed criteria shall be settled to the satisfaction of the Government.

3.9 Internal voids. The billet shall be free from internal voids, porosity, inclusions or contaminants larger than one-eighth (1/8) inch in any direction.

3.10 Surface quality of top of billet. The top of the billet shall be free from evidence of end cracks or shrinkage voids after top cropping.

3.11 Plate producibility. Plate rolled from the first article billets shall meet the following requirements when processed using qualified production procedures and analyzed in accordance with 4.5.1.7:

Hardness Rockwell A	62.0-68.0
Transverse Yield Strength	72,000 psi, minimum
Transverse Tensile Strength	125,000 psi, minimum
Transverse Percent Elongation	20 percent, minimum
Longitudinal Yield Strength	68,000 psi, minimum
Longitudinal Tensile Strength	143,000 psi, minimum
Longitudinal Percent Elongation	15 percent, minimum
Transverse Fracture Toughness	For information

No cracks
No holes
No surface discontinuities (e.g., indentations) greater than 0.015 inch in depth and 1.0 inch in length.
Thickness uniformity

3.12 Manufacturing methods report/process control documents. The contractor shall generate, maintain and perform all manufacturing operations in accordance with Manufacturing Methods Reports (Process Control Documents). Process Control Documents (PCDs) shall include descriptions of all production

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processes, procedures and equipment (including in-process inspection equipment) used to manufacture and inspect the depleted uranium billet. The descriptions shall identify, as a minimum, the complete sequence of operations and schematic line layouts, basic fabrication equipment (i.e., type, model number, etc.), control parameters and ranges associated with the fabrication process (e.g., time, pressure), associated and special processes, in-process inspections, gages, setups, fixtures and applicable in-process inspection procedures.

Manufacturing Methods Reports/Process Control Documents shall be submitted to, and reviewed and classified (see 3.12.1) by the procuring agency prior to performance of the First Article Tests and Examinations and shall be revised and submitted for review whenever changes are made to the approved manufacturing process. A copy of the latest approved Process Control Documents shall be made available to the Government for periodic process verification. All documents shall be submitted in accordance with DI-P-1604 (see 6.2.1 and 6.2.2.6).

3.12.1 Classification of process control documents. Processes defined by the contractor generated Process Control Documents shall be classified as either Class I or Class II as defined below.

a. Class I - Class I PCDs are, or contain, those processes or processing parameters which either have a major impact upon the safety or performance of the end item; or are not easily or cost effectively verified by inspection or test. Changes to Class I PCDs require technical agency review prior to implementation and will normally require some level of qualification (e.g., an additional first article test or portion thereof) beyond normal quality conformance inspection criteria.

b. Class II - Class II PCDs are all procedures and processing parameters not classified as Class I. Changes made to Class II PCDs must be submitted to the technical agency within five working days of implementation and may require the submittal of an additional first article sample.

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3.12.2 Review of manufacturing methods reports. Government review of Manufacturing Methods Reports shall not be construed as certification of a process to make acceptable product. Government review of Manufacturing Methods Reports is to determine that sufficient information is contained therein and that sufficient control of parameters is exercised to permit the manufacture of a consistent product. Any conflict as to the adequacy of the PCDs shall be settled to the satisfaction of the Government.

3.13 Workmanship. The billet shall be free of any burrs, surface contamination by foreign materials, cold shuts and other material inclusions on the billet surface. The billet packaging shall be free from splinters, burrs or sharp edges which may be injurious to personnel handling the shipping container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample for evaluation in accordance with the sample quantities indicated:

Number of Billets per Casting Lot	Number of Casting Lots
1, 2, or 3	20
4	15
5	12
6	10
7	9
8	8
9	7
10 or more	6

4.3.2 Inspections to be performed. The first article sample shall be subjected to the tests and examinations prescribed by Table I in the quantities specified therein. In addition, the Government reserves the right to subject any assembly, component or test specimen contained in the first article sample to any or all of the requirements contained in the acquisition document or the Technical Data Package.

4.3.3 Rejection. If any assembly, component or test specimen contained in the first article sample fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component or test specimens in the sample to comply with any of the stated requirements.

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Table I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70838 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Billet, Depleted Uranium	1 OF 1	None
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Depleted uranium billet				
	Hydrogen content	(a)		3.5	4.5.1.1
	Chemical composition	(a)		3.4	4.5.1.2
	Billet configuration	(a)		3.7	4.5.1.3
	Matrix oxides	(a)		3.6	4.5.1.6
	Traceability	(a)		3.3	Records
	Top of billet surface quality	(a)		3.10	Visual
	Marking missing or illegible	(a)		3.3.1	Visual
	Internal voids	(a)		3.9	4.5.1.4
	Macrostructure (For information)	(b)		3.2.2	4.5.1.5.1
	Microstructure (For information)	(c)		3.2.3	4.5.1.5.2
	Plate producibility	(d)		3.11	4.5.1.7
	Workmanship	(a)		3.13	Visual

NOTES:
(a) All billets from required number of casting lots.
(b) See 4.5.1.5.1.
(c) See 4.5.1.5.2.
(d) See 4.5.1.7.

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected one hundred percent (100%) to determine compliance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one uninterrupted and unchanged process, using the same materials and methods, in accordance with the same drawings, same revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Inspection lots of depleted uranium billets will be in compliance with the requirements specified herein.

Characteristic	Inspection lot criteria
Chemical composition	Each casting lot.
Hydrogen content	Each casting lot.
All other characteristics	Each billet.

4.4.2 Inspection. The examinations and tests prescribed in the "Quality Conformance Inspection" tables 4.4.2.1 and 4.4.2.2 shall be performed on inspection lots as defined in 4.4.1.

a. Classification of defects. Characteristics (defects) are classified as Critical, Major or Minor in accordance with MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70838 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Billet, Depleted Uranium				None
					NEXT HIGHER ASSEMBLY
					Not Applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Chemical analysis	(a)		3.4	4.5.2.2
102	Hydrogen content	(a)		3.5	4.5.2.1
103	Billet configuration		100%	3.7	Gage
104	Matrix oxides	(b)		3.6	4.5.2.3
105	Traceability		100%	3.3	Records
106	Marking missing or illegible		100%	3.3.1	Visual
107	Subsequent processing capability		100%	3.8	4.5.2.4, (c)
108	Top of billet surface quality		100%	3.10	Visual
Minor					
201	Workmanship		100%	3.13	Visual

NOTES:
(a) One billet per casting lot.
(b) One per fifty casting lots.
(c) The contractor shall establish criteria 7
r assuring the future processability of each billet during performance of the first article and shall apply these criteria during production.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-70838 (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.2	Billet, Depleted Uranium, Packaged	1	None
			NEXT HIGHER ASSEMBLY: Not Applicable

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Surface contamination	(a)		5.4/5.5	4.5.3.1
102	Surface radiation	(a)		5.4/5.5	4.5.3.2
103	Transport index	(a)		5.4/5.5	4.5.3.3
104	Marking missing, incomplete or illegible	(a)		5.6	Visual
Minor					
201	Workmanship	(a)		3.13	Visual

NOTES:
(a) Each packing/shipping container.

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4.4.3 Testing. Required tests are prescribed in the "Quality Conformance Inspection" 4.4.2.1 and 4.4.2.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/ Inspection Method" column starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs in accordance with DI-R-1714 (see 6.2.1 and 6.7) and the terms of the ordering document.

4.5 Methods of inspection.

4.5.1 First article.

4.5.1.1 Hydrogen content. Each billet from each of the required number of casting lots (see 4.3.1) contained in the first article sample shall be analyzed for hydrogen content. Hydrogen analysis shall be performed by a certified testing laboratory (see 4.5.1.1.1 and 6.8) on a two-gram sample representing the top of the billet after cropping (the sample may be taken from the bottom of the top crop). Any billet failing to meet the requirements of 3.5 upon initial hydrogen content determination may be reworked (see 6.2.e and 6.14). Failure of the hydrogen content for any billet within a casting lot to be within ± 0.2 parts per million (ppm) of the average hydrogen content for that casting lot shall be cause for rejection of the first article sample. Results of hydrogen content testing of the first article sample shall be submitted to the Government in accordance with UDI-R-22706A/T (see 6.2.1).

4.5.1.1.1 Certification for hydrogen content testing. Prior to performing any hydrogen content analyses for acceptance of depleted uranium billets, the testing laboratory must be certified by the technical agency.(see 4.4.4 and 6.8).

4.5.1.2 Chemical analysis. Each billet from each casting lot contained in the first article sample shall be analyzed for the content of the impurities specified by 3.4. Analyses shall be performed using standard laboratory procedures and equipment as approved by the Government (see 4.4.4) using samples representing the top of the billet after cropping (the sample may be taken from the bottom of the top crop). Determination of the weight percent for total uranium content may be performed by

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difference (i.e., one hundred percent (100%) minus the cumulative total of all other elements). Results of chemical analysis testing of the first article sample shall be submitted to the Government in accordance with UDI-R-22706A/T (see 6.2.1). The first article sample shall be rejected if the chemical analyses of the first article sample fails to meet either of the following criteria:

a. Each billet submitted as part of the first article sample shall meet the requirements of 3.4;

b. The range in content for each element within each casting lot shall not exceed twenty percent of the allowed maximum content for that element.

4.5.1.3 Billet configuration. Each billet contained in the first article sample shall be inspected for each of the billet dimensions or physical features specified in the purchase order or contract using inspection equipment reviewed by the Government in accordance with 4.4.4.

4.5.1.4 Ultrasonic test. Each billet from each of the casting lots contained in the first article sample shall be subjected to ultrasonic testing using test equipment and procedures reviewed in accordance with 4.4.4. Copies of all ultrasonic tracings from the first article sample shall be submitted to the Government in accordance with UDI-R-22706A/T (see 6.2.1). If any billet displays an ultrasonic indication greater than that allowed by 3.9 the first article sample shall be rejected.

4.5.1.5 Metallography. After completion of the ultrasonic testing, six billets contained in the first article sample shall be selected by the Government for metallographic examination in accordance with 4.5.1.5.1 and 4.5.1.5.2. Results of the metallographical examinations shall be for information only.

4.5.1.5.1 Macrostructure. Macrostructure specimens shall be fabricated in each of three orientations (one transverse and two longitudinal) from two billets for each orientation using standard laboratory procedures (see 4.4.4). Required orientations are: top view (transverse, see 4.5.1.5.1.1); front view (longitudinal 1, see 4.5.1.5.1.2); and side (end) view (longitudinal 2, see 4.5.1.5.1.3).

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4.5.1.5.1.1 Transverse macrostructure. Transverse macrostructure specimens shall be prepared within one-fourth of an inch (0.025-inch) of the following locations: the top of the billet; the length centerline of the billet; and two inches from the bottom of the billet. Transverse macrostructure specimens shall be full billet width and depth at specimen location. Billets to be used for transverse macrostructure examination shall be so designated by the Government. Photomacrographs shall be taken at 1X and submitted to the Government in accordance with UDI-R-22706A (see 6.2.1).

4.5.1.5.1.2 Longitudinal 1 macrostructure. Longitudinal 1 macrostructure specimens shall be prepared within one-fourth of an inch (0.025-inch) of the following locations: one inch from each side of the billet and the width centerline of the billet at each the front face, the depth centerline (± 0.250-inch) and the back face of the billet. Longitudinal 1 macrostructure specimens shall be full billet length by two-inches at each specimen location. Billets to be used for longitudinal 1 macrostructure examination shall be so designated by the Government. Photomacrographs shall be taken at 1X and submitted to the Government in accordance with UDI-R-22706A (see 6.2.1).

4.5.1.5.1.2 Longitudinal 2 macrostructure. Longitudinal 2 macrostructure specimens shall be prepared within one-fourth of an inch (0.025-inch) of the following locations: one inch from each side of the billet and the width centerline of the billet. Longitudinal 2 macrostructure specimens shall be full billet length by full billet depth at each specimen location. Billets to be used for longitudinal 2 macrostructure examination shall be so designated by the Government. Photomacrographs shall be taken at 1X and submitted to the Government in accordance with UDI-R-22706A (see 6.2.1).

4.5.1.5.2 Microstructure. Microstructure specimens shall be fabricated in each of three orientations (two longitudinal and one transverse) using standard laboratory procedures (see 4.4.4) from two of the billets selected per 4.5.1.5. Required orientations are: top view (transverse); front view (longitudinal 1); and side (end) view (longitudinal 2). The locations for microstructure examination (all three orientations at each location for each billet) shall be the intersection points of the transverse, longitudinal 1 and longitudinal 2 macrostructure specimens (twenty-seven locations per billet). Billets to be subjected to microstructure examination shall be designated for that purpose by the Government. Photomicrographs shall be taken at 100X and submitted to the Government in accordance with UDI-R-22706A/T (see 6.2.1).

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4.5.1.6 Matrix inclusions. A sample shall be metallographically prepared using standard laboratory procedures from the bottom of the top crop of each billet contained in the first article sample. These samples shall be viewed at 100X under bright field illumination and a photomicrograph taken of each sample. Photomicrographs shall be taken of the region of the sample containing the highest density of inclusions. The photomicrographs will be examined to determine compliance with 3.6. A copy of each photomicrograph from the first article sample shall be submitted to the Government in accordance with UDI-R-22706A.

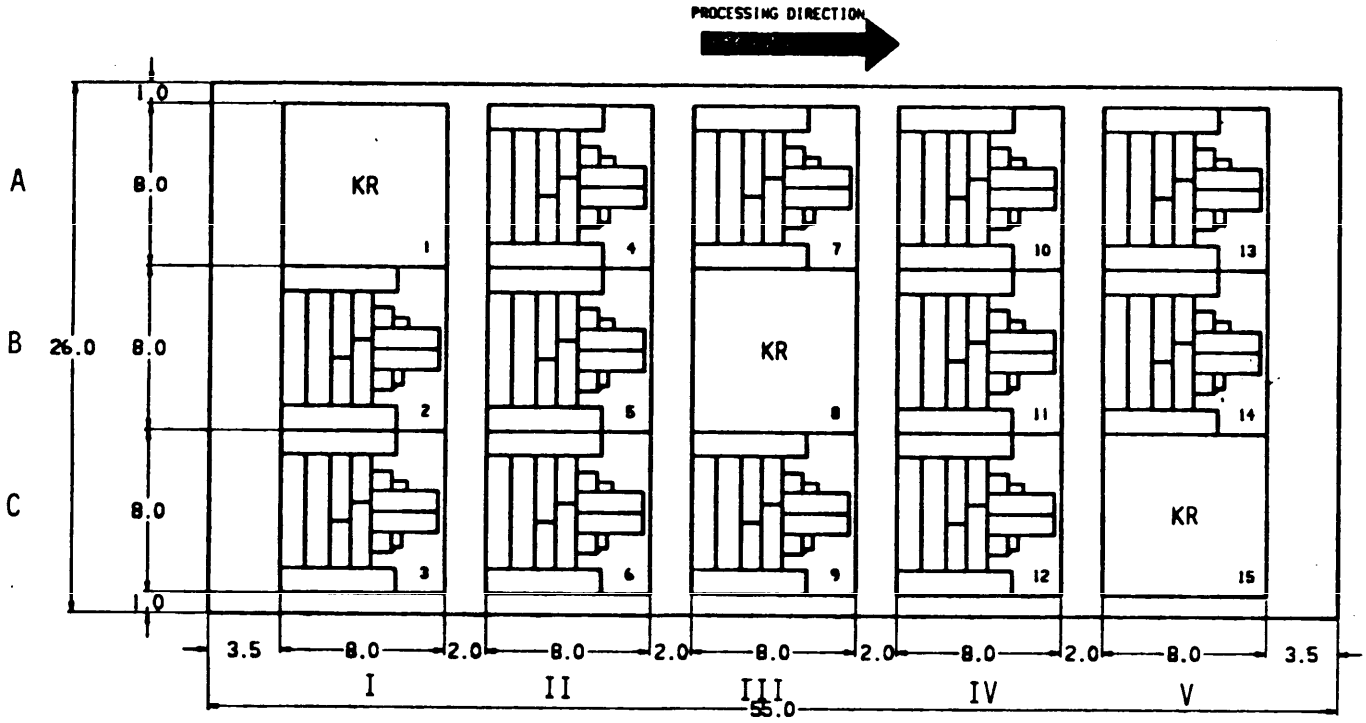
4.5.1.7 Plate producibility. With the exception of the billets selected for metallographic examination, each billet contained in the first article sample will be further processed using previously qualified production equipment, procedures and personnel.

4.5.1.7.1 Surface examination. The plates rolled from the first article sample shall be visually examined for the defects cited in 3.7. If more than five percent (5%) of the plates rolled from the first article sample exhibit rejectable defects (see 3.7) the first article shall be rejected.

4.5.1.7.2 Uniformity study. Twelve plates rolled from the first article sample shall be selected by the Government and tested to determine the uniformity of the plate and compliance with 3.11. The twelve plates will be sectioned in accordance with Figure I. Each section shall be identified by parent plate serial number and location within that plate (see Figure I for section numbering). In addition, four plates produced from billets from a previously qualified supplier and rolled during the same timespan as the first article billets shall be selected and used as controls for the uniformity study (i.e., the four plates produced from the qualified source billets shall be analyzed for compliance with the requirements specified herein). If the first article sample fails to meet the requirements herein and the control plates meet the requirements, the first article will be rejected. If however, both the first article sample and the control plates fail to meet the requirements specified herein, the uniformity study portion of the first article will be considered a "no-test" and the uniformity study will be repeated. If the first article sample meets the requirements specified herein and the control sample fails, the first article will be considered valid.

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FIGURE I



SCALE: 1/4" - 1"

NOTES:

1. 3 WINDOWS - ONE KR SPECIMEN PER WINDOW. NOTCH WILL BE PARALLEL TO ROLLING DIRECTION.

2. 12 WINDOWS - TWO TRANSVERSE TENSILE SPECIMENS
   TWO LONGITUDINAL TENSILE SPECIMENS
   ONE HARDNESS SPECIMEN FOR ROCKWELL A
   TWO TRANSVERSE DROP TOWER SPECIMENS
   TWO LONGITUDINAL DROP TOWER SPECIMENS

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4.5.1.7.2.1 Drop tower. Transverse and longitudinal drop tower specimens will be fabricated from each of the twelve plates in accordance with Figures III and IV respectively from the locations prescribed by Figures I (sections) and II (locations within sections). Specimens shall be identified by orientation (longitudinal or transverse), section and plate serial number. Results of drop tower testing are for information and will not determine the disposition of the first article sample.

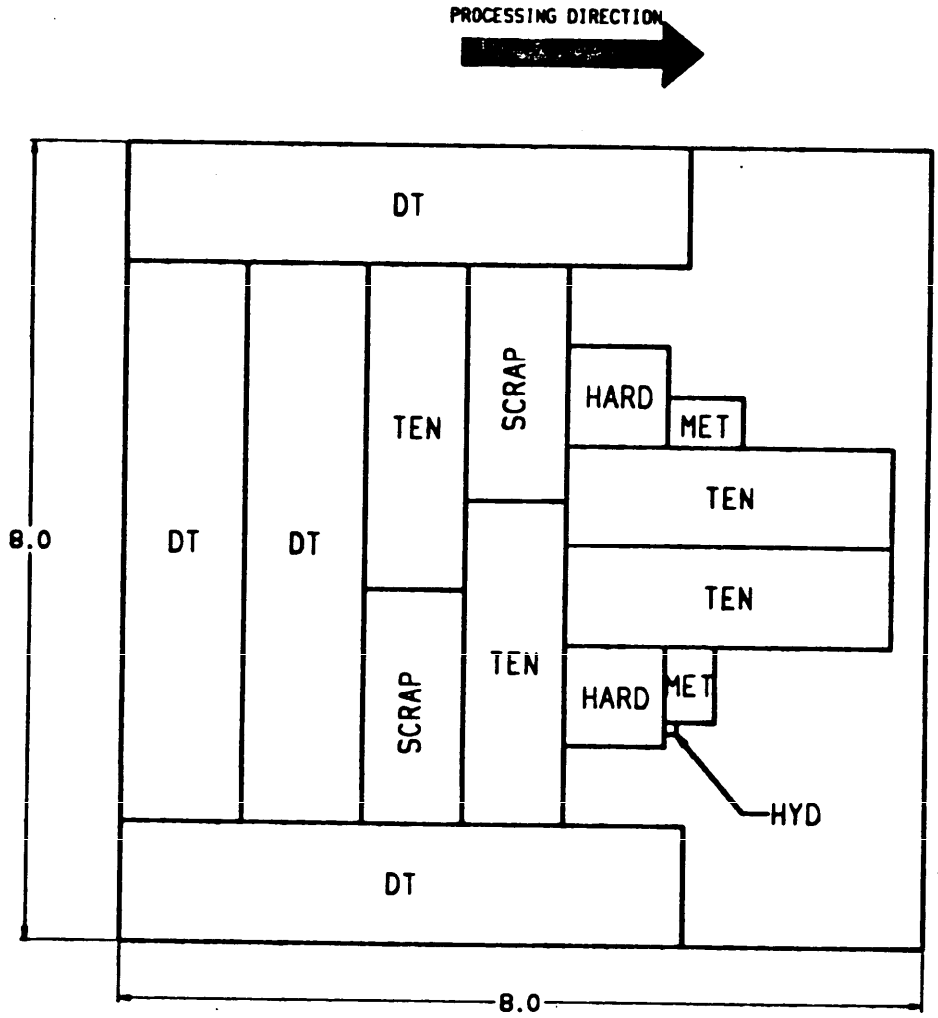
4.5.1.7.2.2 Fracture toughness. Fracture toughness (K_r) specimens will be fabricated from each of the twelve plates in accordance with Figure V from the locations prescribed by Figure I. Specimens shall be identified by section and plate serial number. Results of fracture toughness testing shall be for information and will not determine the disposition of the first article sample.

4.5.1.7.2.3 Tensile properties. Transverse and longitudinal tensile specimens will be fabricated from each of the twelve plates in accordance with Figures VI and VII, respectively, from the locations prescribed by Figures I (sections) and II (locations within sections). Specimens shall be identified by orientation (longitudinal or transverse), section and plate. Tensile tests shall be performed on the as-produced thickness in accordance with ASTM E 8 using a machine cross-head speed of 0.05 inch per minute to material yield and then 0.5 inch per minute to material failure. Test results shall be analyzed for ultimate tensile strength, yield strength and percent elongation shall be traceable to plate and location identification. Any test specimen failing to meet the requirements of 3.11 shall be examined to determine the validity of the test and test results using the criteria of ASTM E 8. Any dispute as to the validity of a test shall be resolved to the Government's satisfaction.

4.5.1.7.2.3.1 Yield strength. Yield strength for tensile testing of the plate shall be determined using a 0.65 percent extension (see 6.10).

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FIGURE II

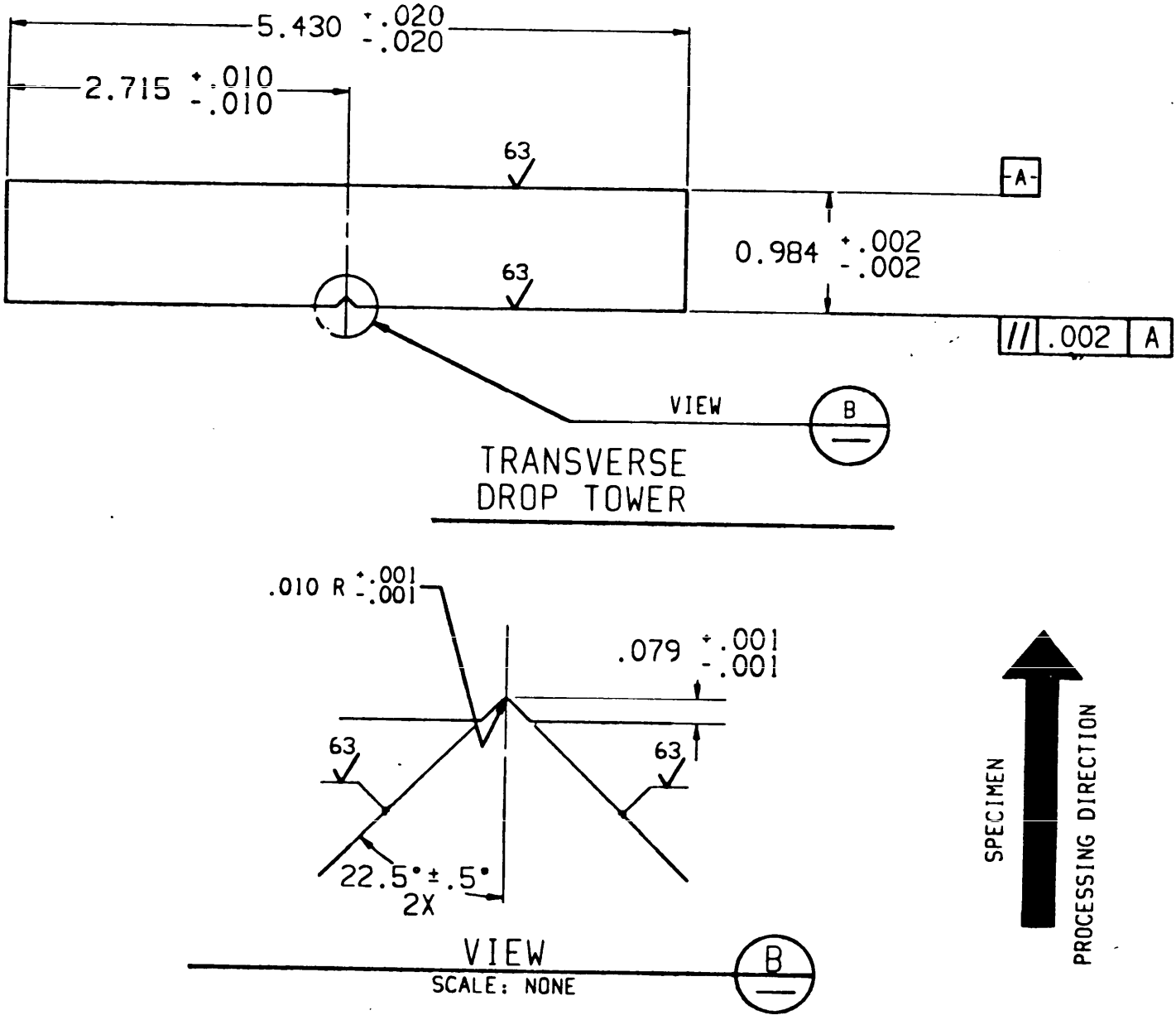


SCALE: FULL

	ACTUAL SIZE	SHEAR SIZE
HARD	1.0 X 1.0	1.0 X 1.0
DT	5.43 X .984	5.6 X 1.2
TEN	3.0 X .75	3.25 X 1.0
MET	.50 X .75	.50 X .75
HYD	.125 X .125	.125 X .125

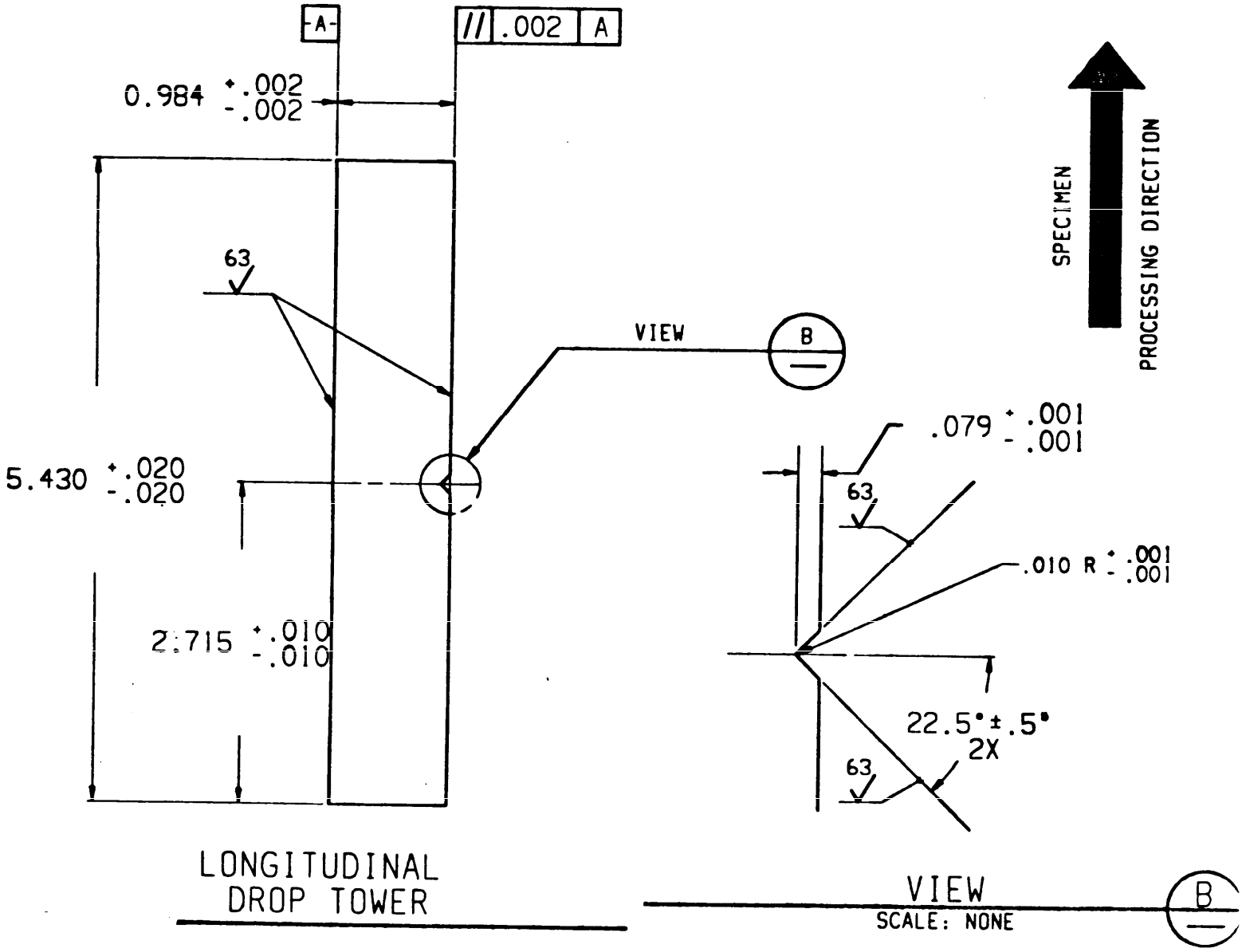
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FIGURE III



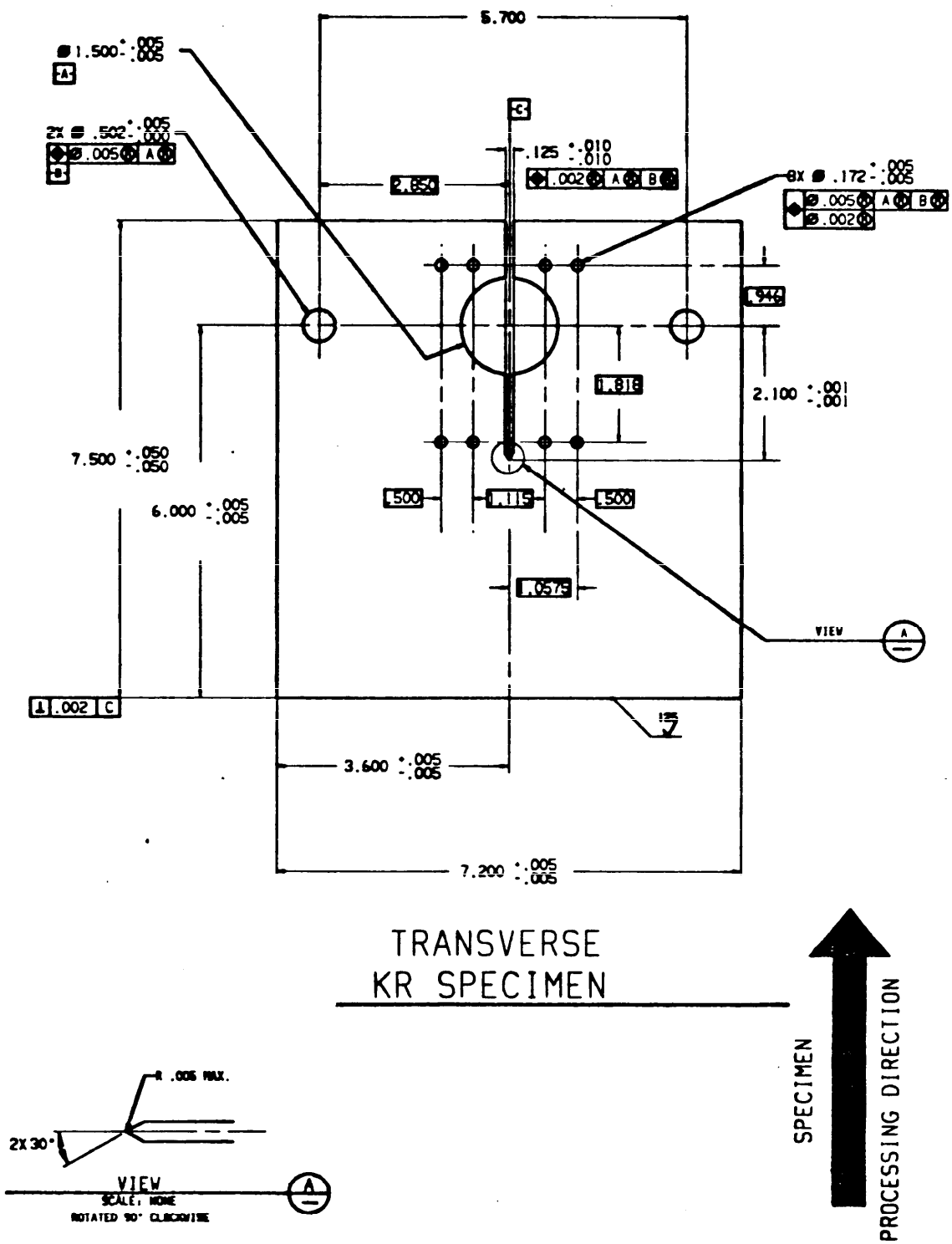
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FIGURE IV



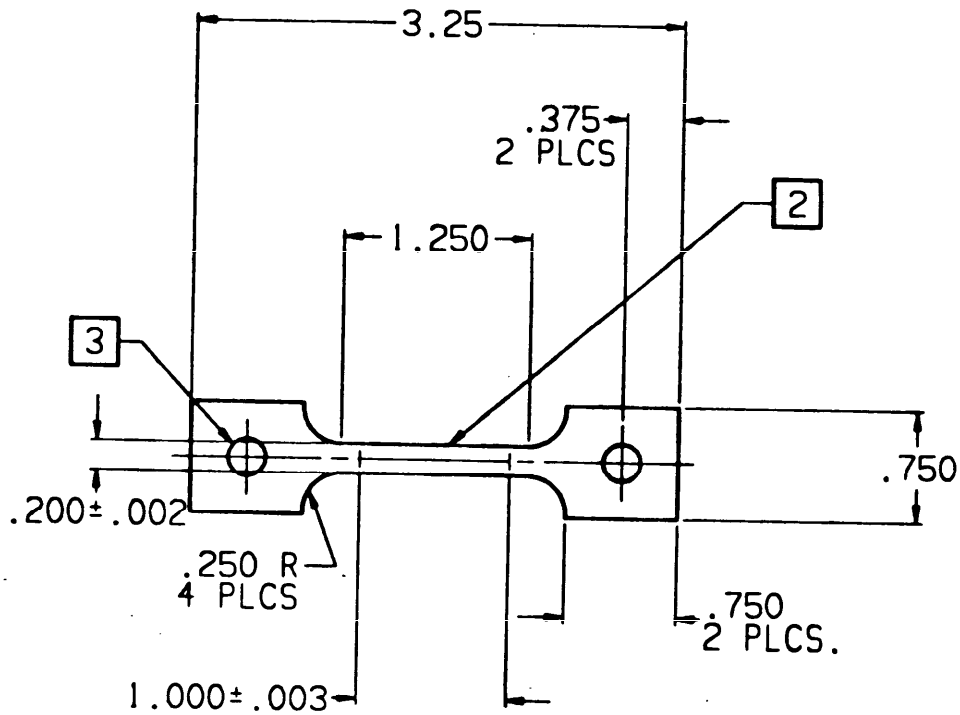
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FIGURE V



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FIGURE VI



TRANSVERSE TENSILE

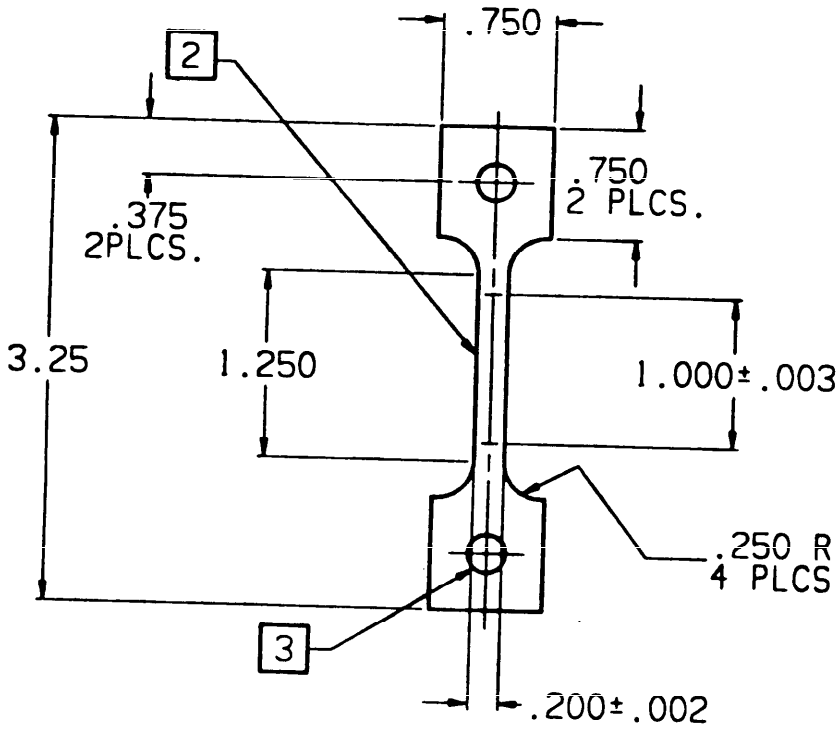
1. SPECIMEN THICKNESS SHALL BE FULL GAGE THICKNESS OF PLATE BEING TESTED.

2. MAXIMUM GAGE WIDTH MUST BE WITHIN 0.001 INCH AT ENDS OF THE REDUCED SECTION. TAPER OF GAGE WIDTH TO CENTER OF SPECIMEN IS ALLOWED BUT CANNOT REDUCE WIDTH BY MORE THAN 0.003 INCH.

3. DIAMETER 0.252 + .005/-.000 HOLE MUST BE ON CENTER LINE OF REDUCED SECTION WITHIN 0.002 INCH.

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FIGURE VII



LONGITUDINAL TENSILE

1. SPECIMEN THICKNESS SHALL BE FULL GAGE THICKNESS OF PLATE BEING TESTED.

2. MAXIMUM GAGE WIDTH MUST BE WITHIN 0.001 INCH AT ENDS OF THE REDUCED SECTION. TAPER OF GAGE WIDTH TO CENTER OF SPECIMEN IS ALLOWED BUT CANNOT REDUCE WIDTH BY MORE THAN 0.003 INCH.

3. DIAMETER 0.252 + .005/-.000 HOLE MUST BE ON CENTERLINE OF REDUCED SECTION WITHIN 0.002 INCH.

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4.5.1.7.2.3.2 Acceptance. Acceptance of the first article sample for tensile properties shall be based upon the following criteria:

a. All valid (non-disregarded) test results for each analysis (tensile strength, yield strength and percent elongation) in each alignment (longitudinal and transverse) for each plate shall meet the requirements of 3.11;

b. The average (mean) of all valid test results (all plates) for each analysis in each alignment minus two standard deviations (see 6.11) for that analysis in that alignment for all valid test results shall meet the requirements of 3.11;

c. Tensile properties in each orientation shall be uniform throughout the entire plate as determined by a modified Wilcoxon Rank-Sum positional analysis (see 6.12). If the absolute difference of rank sums exceeds the critical value for any pair of columns (critical value twenty-one) or rows (critical value twelve), the first article shall be rejected.

If the first article sample fails to meet any of the criteria specified by a., b. or c., the first article sample shall be rejected.

4.5.1.7.2.3.3 Hardness. Hardness measurements shall be taken in accordance with ASTM E 18 using the Rockwell A scale at the locations prescribed by Figures I (sections) and II (location within sections) for all twelve plates. Three hardness measurements shall be taken and averaged at each required location. Failure of any average hardness for any required section to meet the requirement of 3.11 shall result in the rejection of the first article sample.

4.5.1.7.3 Thickness uniformity. Each plate produced from the first article billet sample shall be inspected for thickness using automated inspection equipment. Thickness measurements shall be taken at one-inch (maximum) intervals along three tracks in the rolling direction. These tracks shall be equidistantly located, with the outer two tracks approximately five inches from either longitudinal edge of the plate. Each track from each plate shall be divided into discrete five-inch long segments for analysis purposes only. The ability of billets to produce plates of a uniform thickness shall be determined on a twelve plate sample selected by the Government using a multi-level decision plan as follows:

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a. If all points within a plate are within the upper and lower thickness limits established for the plate, that plate is acceptable and no further analysis is required for that plate.

b. Within each row of each plate, up to one point may be up to 0.002 inch less than the lower limit established and up to two points, from different five-inch segments, may be up to 0.002 inch greater than the upper thickness limit established for the plate. Any plate meeting this criteria shall be deemed acceptable and no further analysis is required for that plate.

c. The average for each five-inch segment shall be within 0.003 inch of the grand average thickness of the plate. Any plate meeting this criteria shall be deemed acceptable and no further analysis is required for that plate.

d. A one-way Analysis of Variance (ANOVA) test shall be performed on any remaining plates in accordance with 6.13. Any plate for which the "null hypothesis" is rejected (see 6.13) shall be considered a no-test and shall not be cause for rejection of the first article sample.

If any of the twelve plates selected for thickness uniformity determination is not acceptable by any of a. through c., and is not considered a no-test by d., the first article shall be rejected.

4.5.2 Quality conformance inspection.

4.5.2.1 Hydrogen content. One billet from each casting lot shall be analyzed for hydrogen content. Hydrogen analysis shall be performed by a certified testing laboratory (see 4.5.2.1.1 and 6.8) on a two-gram sample representing the top of the billet after cropping (the sample may be taken from the bottom of the top crop). Failure of any hydrogen content analysis to meet the requirement of 3.5 shall result in the rejection of the casting lot represented by that analysis.

4.5.2.1.1 Certification for hydrogen content testing. Prior to performing any hydrogen content of depleted uranium analyses for the Abrams Heavy Armor program, the testing laboratory must be certified by the technical agency (see 4.4.4 and 6.8).

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4.5.2.2 Chemical analysis. One billet from each casting lot shall be analyzed for the content of the impurities specified by 3.4. Analyses shall be performed using standard laboratory procedures and equipment as reviewed by the Government (see 4.4.4) using samples representing the top of the billet after cropping (the sample may be taken from the bottom of the top crop). If billets are gang-cast, the chemical analysis shall be performed on the billet from which the shortest top-crop was taken. If the billets are sequentially cast, chemical analysis shall be performed on the last billet poured. Failure of the chemical composition of any analysis to meet the requirements of 3.4 shall result in the rejection of the casting lot represented by that analysis. Determination of the weight percent for total uranium content may be performed by difference (i.e., one hundred percent (100%) minus the cumulative total of all other elements).

4.5.2.3 Matrix inclusions. A sample shall be metallographically prepared using standard laboratory procedures reviewed in accordance with 3.11 and 4.4.4 from the bottom of the top crop from one billet selected from each fifty casting lots. These samples shall be viewed at 100X under bright field illumination and a photomicrograph taken of each sample. Photomicrographs shall be taken of the region of the sample containing the highest density of inclusions. The photo-micrographs will be examined to determine compliance with 3.6. A copy of each photomicrograph shall be submitted to the Government in accordance with UDI-R-22706A (see 6.2.1).

4.5.2.4 Subsequent processing capability. During performance of the first article the contractor shall establish his criteria for assuring the future processability of the billets. During production the same criteria shall be applied for the acceptance of billets as was used during the first article.

4.5.3 Inspection of packaging.

4.5.3.1 Surface contamination. Each packing/shipping container shall be tested for removable surface contamination on each of its six main surfaces (i.e., top, bottom, front, back, right end and left end). Using a separate clean, dry piece of standard swipe paper for each surface of each container, a one hundred square centimeter (100 cm^2) area on each required surface will be swipe tested. Each test paper shall be identified as to container and applicable surface of that container. The swipe papers shall be analyzed for alpha and beta emissions. Any surface failing to meet the requirements of 5.4 or 5.5 shall be cleaned using procedures reviewed by the Government (see 5.1, 6.2.1 and 6.2.2.9) and retested for removable surface contamination.

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4.5.3.2 Surface radiation. Each packing/shipping container shall be tested for external surface radiation for each of its six main surfaces. Using an Eberline HP-270 probe with accompanying survey meter, or Government approved equivalent (see 4.4.4 and 6.7), each of the six main surfaces shall be scanned for surface radiation to determine compliance with applicable transportation requirements.

4.5.3.3 Transport index. Each packing/shipping container shall be tested for external radiation at a one-meter standoff from the surface for each of its six main surfaces. Using an Eberline HP-270 probe with accompanying survey meter, or Government approved equivalent (see 4.4.4 and 6.7), each of the six main surfaces shall be scanned at an one-meter standoff for external radiation to determine compliance with applicable transportation requirements.

5. PACKAGING

5.1 Cleaning. Any cleaning of the billet or billet packaging/packing shall be performed using processes and agents which will not cause degradation or change in the composition of the billet. The contractor shall submit his proposed cleaning method including a listing of any proposed cleaning agents and reagents to the Government in accordance with DI-P-1604/T (see 6.2.1 and 6.2.2.9).

5.2 Drying. Any drying of the billet shall be performed in such a manner as to not cause degradation of the billet. The contractor shall submit his proposed drying method(s) including a listing of any proposed agents and reagents to be used for drying the billet to the Government in accordance with DI-P-1604/T (see 6.2.1 and 6.2.2.10).

5.3 Preservation. Not applicable.

5.4 Packaging. The level of packaging protection shall be adequate to assure delivery of the billets in a satisfactory condition. In addition, packaging shall meet all Department of Transportation (DOT), Department of Energy (DOE) and Nuclear Regulatory Commission (NRC) requirements as well as the specific requirements of the contractor's depleted uranium handling license. Each unit of packaging shall be tested for surface contamination, external surface radiation and transport index to determine the appropriate level of transportation requirements and marking or placarding of shipments in accordance with Title 49, Code of Federal Regulations (49 CFR), Parts 100 through 177. The contractor shall submit his proposed packaging plan, including any applicable drawings and marking requirements, to the Government in accordance with DI-PACK-80455 (see 6.2.1 and 6.2.2.11).

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5.5 Packing. The level of packing shall be adequate to assure delivery of the billets in a satisfactory condition. In addition, packing of the billets shall meet all Department of Transportation (DOT), Department of Energy (DOE) and Nuclear Regulatory Commission (NRC) requirements as well as the specific requirements of the contractor's depleted uranium handling license. Each unit of packing shall be tested for surface contamination, external surface radiation and transport index to determine the appropriate level of transportation requirements and marking/placarding of shipments in accordance with 49 CFR, Parts 100 through 177. The contractor shall submit his proposed packing plan, including any applicable drawings and marking requirements, to the Government in accordance with DI-PACK-80455 (see 6.2.1 and 6.2.2.11).

5.6 Marking. Marking of shipments and packing containers shall be in accordance with MIL-STD-129 and shall comply with the applicable requirements of 49 CFR as determined by external surface radiation and transport index levels. A packing list shall be included citing the billet serial numbers contained within each container.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The billets covered by this specification are intended for further processing for use in the Abrams Heavy Armor for the M1A1 Abrams Main Battle Tank.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, date and revision of this document.

b. Requirements for first article examination and test (see 3.2, 4.3 and 6.3).

c. Dimensions and configuration of the billets being procured (see 3.7).

d. Higher level quality requirements (MIL-Q-9858 and DI-R-5297).

e. Rework/repair provisions in accordance with 6.14.

f. Data requirements (see 6.2.1).

g. Government furnished material (see 6.5), if any.

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6.2.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by the approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the ordering document. When the provisions of the DOD FAR Supplement, Part 27, Sub-part 27.410-6 are invoked and the DD form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the ordering document requirements. Deliverable data required by this specification is cited in Table II.

6.2.2 Government review authority.

6.2.2.1 Quality program plan. Unless otherwise specified in the acquisition document or the CDRL, the contractor shall submit his Quality Program Plan in accordance with DI-R-5297 (see 6.2.d) to the Government Quality Assurance Representative responsible for his production facility for review.

6.2.2.2 Traceability plan. Unless otherwise specified in the acquisition document, the contractor shall submit his traceability plan (serialization plan) in accordance with DI-R-5297/T (see 3.3) to the following for review:

Office of the Project Manager - Abrams Tank Systems
ATTN: AMCPM-ABMS-S
Warren, MI 48397-5000.

6.2.2.3 Billet marking plan. Unless otherwise specified in the acquisition document, the contractor shall submit his planned method for the marking of billets in accordance with DI-P-1604/T (see 3.3.1) to the following for review:

Office of the Project Manager - Abrams Tank Systems
ATTN: AMCPM-ABMS-S
Warren, MI 48397-5000.

6.2.2.4 Subsequent processing capability criteria. Unless otherwise specified in the acquisition document, the contractor shall submit his criteria for determination of the subsequent processing capability of billets in accordance with DI-R-5297/T (see 3.8) to the following for review:

Office of the Project Manager - Abrams Tank Systems
ATTN: AMCPM-ABMS-Q
Warren, MI 48397-5000.

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TABLE II. Data Requirements

Paragraph	Applicable DID	Title	Recommended Tailoring
3.3	DI-R-5297/T	Quality Program Plan	Traceability plan
3.3.1	DI-P-1604/T	Manufacturing Methods Report	Process Control Documents/Billet marking
3.3.2	UDI-R-22706A	Records, Inspection and Test	None
3.8	DI-R-5297/T	Quality Program Plan	Subsequent processing capability criteria
4.4.4	DI-R-1714	Quality Engineering Acceptance Inspection Equipment Description Documentation	None
3.12	DI-P-1604	Manufacturing Methods Report	Process Control Documents
4.5.1.1/4.5.1.2/ 4.5.1.4/4.5.1.5.1/ 4.5.1.5.2/ 4.5.1.6	UDI-R-22706A/T	Records, Inspection and Test	First article test results for: hydrogen content, chemical analysis, ultrasonic test, microstructure, macrostructure and matrix oxides.
4.5.2.3	UDI-R-22706A	Records, Inspection and Test	Matrix oxide photomicrigraphs
4.5.3.2/ 4.5.3.3	DI-R-1714	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	External surface radiation and transport index determination equipment
5.1	DI-P-1604/T	Manufacturing Methods Report	Process Control Documents/cleaning agents
5.2	DI-P-1604/T	Manufacturing Methods Report	Process Control Documents/drying operations
5.4/5.5	DI-PACK-80455	Packaging Plan	None
6.2.d	DI-R-5297	Quality Program Plan	None
6.14	DI-R-5297/T	Quality Program Plan	Rework procedures
6.14	DI-E-2038	Engineering Change Proposals (ECP's) and Requests for Deviations and Waivers (Short Form)	Repair procedures

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6.2.2.5 Acceptance inspection equipment. Unless otherwise specified in the acquisition document, the contractor shall submit his designs for acceptance inspection equipment in accordance with 6.7 and DI-R-1714 (see 4.4.4) to the following for review:

Office of the Project Manager - Abrams Tank Systems
ATTN: AMCPM-ABMS-Q
Warren, MI 48397-5000.

6.2.2.6 Manufacturing methods report. Unless otherwise specified in the acquisition document, the contractor shall submit his manufacturing methods report (process control documents) in accordance with DI-P-1604 (see 3.12) to the following for review and classification:

Office of the Project Manager - Abrams Tank Systems
ATTN: AMCPM-ABMS-S
Warren, MI 48397-5000.

6.2.2.7 Rework procedures. Unless otherwise specified in the acquisition document, the contractor shall submit his rework procedures, including corrective actions, in accordance with DI-R-5297/T (see 6.14) to the following for review:

Office of the Project Manager - Abrams Tank Systems
ATTN: AMCPM-ABMS-S
Warren, MI 48397-5000.

6.2.2.8 Repair procedures. Unless otherwise specified in the acquisition document, the contractor shall submit repair procedures in accordance with DI-E-2038 (see 6.14) to the following for review and disposition:

Commander
US Army Tank-Automotive Command
ATTN: AMCPM-ABMS
Warren, MI 48397-5000.

or Contracting Officer as designated in the acquisition document.

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6.2.2.9 Cleaning processes. Unless otherwise specified in the acquisition document, the contractor shall submit his proposed cleaning operations, including and agents/reagents used, in accordance with DI-P-1604/T (see 5.1) to the following for review:

Commander
US Amy Tank-Automotive Command
ATTN: AMSTA-CZ
Warren, MI 48397-5000.

6.2.2.10 Drying processes. Unless otherwise specified in the acquisition document, the contractor shall submit his proposed drying operations, including and agents/reagents used, in accordance with DI-P-1604/T (see 5.2) to the following for review:

Commander
US Amy Tank-Automotive Command
ATTN: AMSTA-CZ
Warren, MI 48397-5000.

6.2.2.11 Packaging/Packing plan. Unless otherwise specified in the acquisition document, the contractor shall submit his packaging/packing plan, including applicable drawings, in accordance with DI-PACK-80455 (see 5.4 and 5.5) to the following for review:

Office of the Project Manager - Abrams Tank Systems
ATTN: AMCPM-ABMS
Warren, MI 48397-5000.

6.3 Inspection for first article. When a first article inspection is required by the acquisition document, the item(s) included in the first article sample shall have been produced in accordance with contractual requirements. The first article sample should consist of the required quantity of casting lots of billets. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles.

Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. In addition, provisions for additional first article test(s), or portions thereof, should be made.

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6.4 Definitions. The following definition of terms apply when used in conjunction with this specification.

Billet - A billet is the end product of the casting operation following all cropping and other, if applicable, machining operations and is suitable for further processing.

Casting lot - A casting lot is the sum total of material resulting from the melting and casting of a single furnace charge of depleted uranium.

Recyclable scrap - Recyclable scrap is depleted uranium top croppings, sectioned plates and sectioned billets which are capable of being remelted and processed into billets which meet the requirements specified herein. Swarf and other by-products of production operations not specifically cited above are excluded from the category of recyclable scrap.

Repair - Repair is the reprocessing of nonconforming material in accordance with approved written procedures and operations to reduce, but not completely eliminate, the nonconforming condition (see 6.14).

Rework - Rework is the reprocessing of nonconforming material to make it conform completely to the drawings, specifications and acquisition document requirements (see 6.14).

6.5 Government-furnished and Government-loaned property. The Government, at its option, may provide the contractor with recyclable scrap material (see 3.1 and 6.4) for processing into billets.

6.6 Subject term (key word) listing.

Abrams Heavy Armor
Casting
Ingot
Low Specific Activity
M1A1 Abrams Main Battle Tank

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6.7 Inspection equipment designs. Inspection equipment designs are of two types, Government Special Inspection Equipment (SIE) and contractor designs. SIE designs, if applicable, are designated by drawing numbers under the "Method of Inspection" heading in the Classification of Defects and Tests Tables contained in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Unless otherwise specified, the contractor may submit alternate or modified contractor equipment for external surface radiation testing (4.5.3.2) and determination of transport index (4.5.3.3).

6.7.1 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment (i.e., unmodified equipment which is cataloged and available for purchase by the general public) which the contractor proposes to use. Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data.

Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby. Designs shall be of the category and form (per DOD-D-1000) specified in the CDRL (DD Form 1423). The item model number, component or assembly drawing number, and a narrative description of the defect characteristic (see Quality Conformance Inspection paragraphs 4.4.2.1 and 4.4.2.2) shall be shown on each contractor design.

6.8 Hydrogen testing certification. Unless otherwise specified in the acquisition document, certification of laboratories for the determination of hydrogen in depleted uranium will be performed under the auspices of the agency designated for approval of acceptance inspection equipment designs. Certification of testing laboratories will be accomplished in the following manner:

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a. The Government will provide the testing laboratory with depleted uranium test standards at four different hydrogen levels.

b. The laboratory shall calibrate their testing equipment in accordance with their approved acceptance inspection equipment operating procedures.

c. Five valid hydrogen analyses shall be made for each of the hydrogen level standards provided. A maximum of four total non-valid (disregarded) test results due to test methodology, preparation or equipment are allowed.

d. The average of the five analyses at each level must be within $\pm$ 0.10 ppm of the "standard" hydrogen content for that level (no rounding of average is authorized). In addition, the rounded value for each valid test analysis must be within $\pm$ 0.2 ppm of the "standard" hydrogen content for that coupon. Both of these criteria must be met for testing laboratory certification.

6.9 Handling of depleted uranium. Care shall be taken to ensure that all procedures for the handling of depleted uranium as outlined by the Nuclear Regulatory Commission (NRC), Department of Energy (DOE) and the individual contractor's regulations and licenses are followed.

6.10 Determination of yield strength. Yield strength of depleted uranium shall be determined using the 0.65% extension method. Using a load versus displacement plot representing the tension test and with displacement plotted on the x-axis, move 0.0065 inches along the x-axis (for an one-inch extensometer) from the zero point and determine the load (intersection of plot taken parallel to the y-axis) for this displacement. The load value determined provides the force value to be used in the calculation of 0.65% extension yield strength.

6.11 Determination of standard deviation. The standard deviation of a sample is calculated in accordance with the following equation found in AMCP 706-110:

$$\text{Standard Deviation} = \sqrt{\frac{N\Sigma X_i^2 - (\Sigma X_i)^2}{N(N-1)}}$$

Where N = Number of samples and X_i = Value for each test result.

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6.12 Modified Wilcoxon Rank-Sum positional analysis. The test results for each test in each orientation for each column (I through V, see Figure I) of each plate shall be averaged (a minimum of one valid test result for each section is required. If both sets of test results for an orientation within a section are invalid, a double retest shall be performed for that orientation). The average value for each test in each orientation for each column shall be ranked within each plate with a rank of "1" being assigned to the minimum column average and a rank of "5" being assigned to the maximum column average for that plate (in the event of two or more values being identical within a plate, the average rank of all columns with that value shall be used as the rank for each of those columns). For each column I through V, the ranks of that column from each plate will be summed and the absolute difference of the sums for each permutation of two columns shall be computed (see Table III). The absolute value for each pair of differences of column rank-sums shall be compared with the applicable critical value. The critical value for the first article sample of twelve plates with five columns is twenty-one; the critical value for the control plates (four plates with five columns) is twelve. In addition, the test results for each test in each orientation for each row (A, B, C) of each plate shall be averaged. The average value for each test in each orientation for each row shall be ranked within each plate with a rank of "1" being assigned to the minimum row average and a rank of "3" being assigned to the maximum row average for that plate (in the event of two or more row averages being identical within a plate, the average rank of all rows with that value shall be used as the rank for each of those rows). For each row A through C (see Figure I), the ranks of that row from each plate will be summed and the absolute difference of the sums for each permutation of two rows shall be computed (see Table IV). The absolute value for each pair of differences of row rank-sums shall be compared with the applicable critical value. The critical value for the first article sample of twelve plates with three rows is twelve; the critical value for the control plates (four plates with three rows) is seven. The critical values contained herein are based upon the number of plates and the number of columns or rows cited, if any of these parameters are changed, different critical values will be assigned by the Procuring Contracting Officer.

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TABLE III - MATRIX FOR COLUMN COMPARISONS

COLUMN	I	II	III	IV	V
I	-	*	*	*	*
II	-	-	*	*	*
III	-	-	-	*	*
IV	-	-	-	-	*
V	-	-	-	-	-

* Absolute value of difference in rank sums of column averages required.

TABLE IV - MATRIX FOR ROW COMPARISONS

ROW	A	B	C
A	-	*	*
B	-	-	*
C	-	-	-

* Absolute value of difference in rank sums of row averages required.

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6.12.1 Example of modified Wilcoxon rank sum determination. Using the Longitudinal percent elongation data contained in Table V, the following example of a Wilcoxon rank sum analysis for columns for a control sample of plates is provided.

TABLE V
Longitudinal Percent Elongation
(Column Average)

	Column				
Plate	I	II	III	IV	V
1	15.5	26.0	21.3	17.0	26.0
2	18.3	23.5	25.0	21.3	24.8
3	21.0	22.7	23.5	20.7	25.3
4	19.3	22.0	27.5	21.7	28.0

Using the data from Table V, column ranks for each plate are assigned and rank sums for each column are determined as per Table VI.

TABLE VI
Longitudinal Percent Elongation
Column Ranks and Rank Sums

	Column				
Plate	I	II	III	IV	V
1	1	4.5	3	2	4.5
2	1	3	5	2	4
3	2	3	4	1	5
4	1	3	4	2	5
Rank Sum	5	13.5	16	7	18.5

Using Table III, the following absolute differences of rank sums are obtained:

COLUMN	I	II	III	IV	V
I	-	8.5	11	2	13.5
II	-	-	2.5	6.5	5
III	-	-	-	9	2.5
IV	-	-	-	-	11.5
V	-	-	-	-	-

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Comparing the differences of rank sums with the critical value for four plates with five columns (twelve (12)) (see 6.12) it is determined that there is a significant difference in longitudinal percent elongation between columns I and V for the control plates in this example.

6.13 One-way Analysis of Variance for thickness. The thickness data generated in 4.5.1.7.3 shall be analyzed for each plate as follows:

a. The thickness measurements for each track shall be summed, i.e. three sums of thicknesses shall be obtained.

b. The sum of measurements for each track (a.) shall be squared.

c. The values obtained in b. shall each be divided by the number of thickness measurements in each track.

d. The three values obtained in c. shall be summed.

e. The sum of all thickness measurements for the plate shall be taken.

f. The value obtained in e. shall be squared.

g. The value obtained in f. shall be divided by the total number of thickness measurements for that plate.

h. The value obtained in g. shall be subtracted from the value obtained in d. The resultant value is the Sum of the Squares (SS) for the rows.

i. The Sum of the Squares for the means (h.) shall be divided by the degrees of freedom (df) for the rows which is two (2), i.e. the number of rows (3) minus one (1). The resultant value is the Mean Square of the rows.

j. Each thickness measurement for the plate shall be squared.

k. The values obtained in j. shall be summed.

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l. The value obtained in d. shall be subtracted from the value obtained in k. The resultant value is the SS for within the Plate.

m. The degrees of freedom for the SS of the Plate shall be determined. This value is equal to the total number of thickness measurements for the plate minus the number of rows (3).

n. The SS of the Plate (l.) shall be divided by the degrees of freedom for the Plate. The resultant value is the Mean Square for within the Plate.

o. The Means Square for the rows (i.) shall be divided by the Means Square for the Plate (n.) and a resultant "F" value obtained for each plate.

p. The F value obtained for each plate (o.) shall be compared with 3.05. If the F value exceeds 3.05 the null hypothesis (i.e., variation of thickness within the plate is random) shall be rejected for that plate.

6.13.1 Formulas for ANOVA. The following formulas apply for the computation used for determination of the F value for each plate:

	SS	df	MS	F value
Means (Rows)	$\sum \frac{R_i^2}{n_i} - \frac{T^2}{N}$	k - 1	$\frac{SS}{df}$	
Within (Plates)	$\sum x_i^2 - \sum \frac{R_i^2}{n_i}$	N - k	$\frac{SS}{df}$	
Total				$\frac{MS\ (Rows)}{MS\ (Plates)}$

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6.13.2 Abbreviations for ANOVA formulas. The following abbreviations are used in the formulas for the computation of analysis of variance:

SS = the Sum of the Squares (either means (rows) or within (plates).

df = the degrees of freedom for the Sum of the Squares.

MS = the Mean Square (either means (rows) or within (plates).

R = the sum of thickness measurements within a row (track).

n = the number of measurements in a row (track).

T = the sum of all thickness measurements within the plate.

N = the total number of thickness measurements within the plate.

k = the number of rows (tracks).

x = each individual thickness measurement within the plate.

6.14 Rework/repair of nonconforming material. The contractor shall establish and maintain a system which shall identify, segregate (or control if segregation is not practical) and properly dispose of nonconforming material and shall ensure that cost-effective, positive corrective action is taken to prevent, minimize or eliminate nonconformances. All nonconforming material, including in-process material, shall be subject to one of the following procedures:

a. Scrap: Material to be scrapped will be segregated or identified to preclude inadvertent submission to the Government of such material. Adequate records shall be kept to allow for efficient retrieval and summarization. Scrap material shall be disposed of in such a manner as to preclude subsequent resubmission or use by unauthorized persons, agencies, etc. or in accordance with the contract or purchase order.

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b. Rework: Rework (see 6.4) procedures are to be documented and submitted as an inspection system change to the Government in accordance with DI-R-5297/T for review prior to implementation and are subject to the Government's disapproval. In the event operations or processes are to be used to rework nonconforming material other than those operations used to produce the accepted first article (if applicable) or the processes and procedures used to produce the contract quantity, the proposed rework procedure will be submitted to the Procuring Contracting Officer for review prior to implementation. Rework procedures shall allow for maintaining product homogeneity (see 4.4.1).

c. Repair: Repair (see 6.4) procedures are to be documented and submitted as a Request for Deviation/Waiver in accordance with DI-E-2038 to the Procuring Contracting Officer for review and approval prior to implementation.

Whenever the contractor submits a proposed repair or rework procedure or a request for waiver, the submission shall include a description of the nonconformance and a description of the actions taken or to be taken to prevent recurrence. Whenever nonconforming material is subjected to rework procedures the contractor shall verify that all such material is verified or reinspected for all characteristics which may have been affected by the rework procedure.

Proposed repair procedures shall also contain a provision for the reinspection of any variance that may be introduced as a result of the repair procedure, as well as providing assurance to the Government that defective items have been removed.

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